



Under the Sand of Exoplanet Analogs

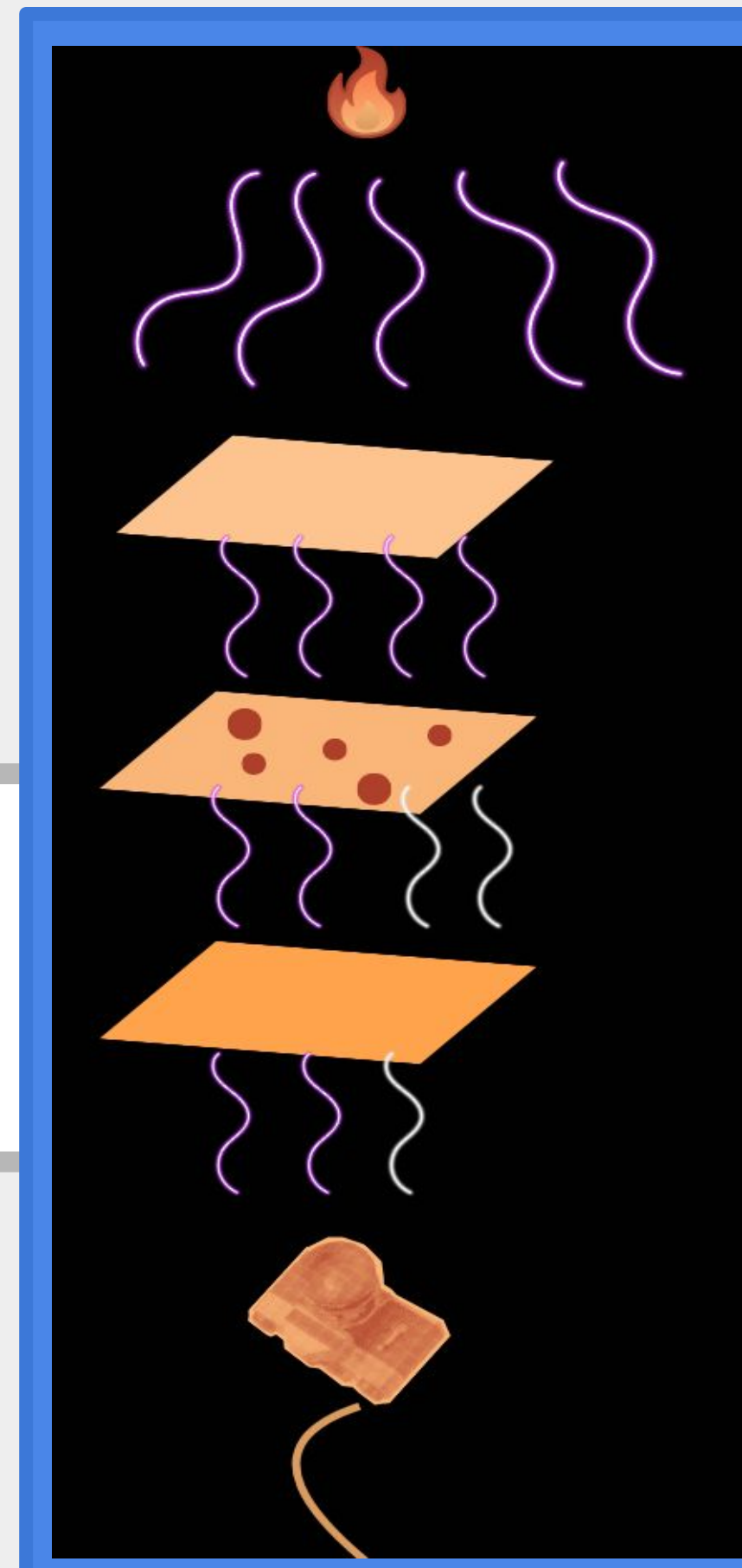
Jannatul Urmi¹, Luca D'Entrone², Timothy Ebegbe³, Genaro Suárez⁴
¹East Bronx Academy, ²Saint Ann's School, ³Brooklyn Technical High School, ⁴AMNH



Introduction

Brown dwarfs hold the key to understanding the properties of atmospheres, including our own. To learn about these warm, large, gas giants, scientists use telescopes to capture the light they emit (Kirkpatrick 2005). This light emitted at several lengths (*wavelengths*) is called *spectrum*. The spectra is filtered through the planet's atmospheres, blocking specific wavelengths which reveals their chemical composition.

Our goal is to determine the relationship between the presence of sand clouds to other gas molecules.



Methods

Using Python through the Jupyter Notebooks software, we identify the presence of specific molecules such as:

- ❖ Water (H_2O)
- ❖ Carbon monoxide (CO)
- ❖ Carbon dioxide (CO_2)
- ❖ Methane (CH_4)
- ❖ Silicates (sand)

To quantify the depth of the signatures, we used the open source python package SEDA (Suárez et al. 2026) to calculate spectral indices.

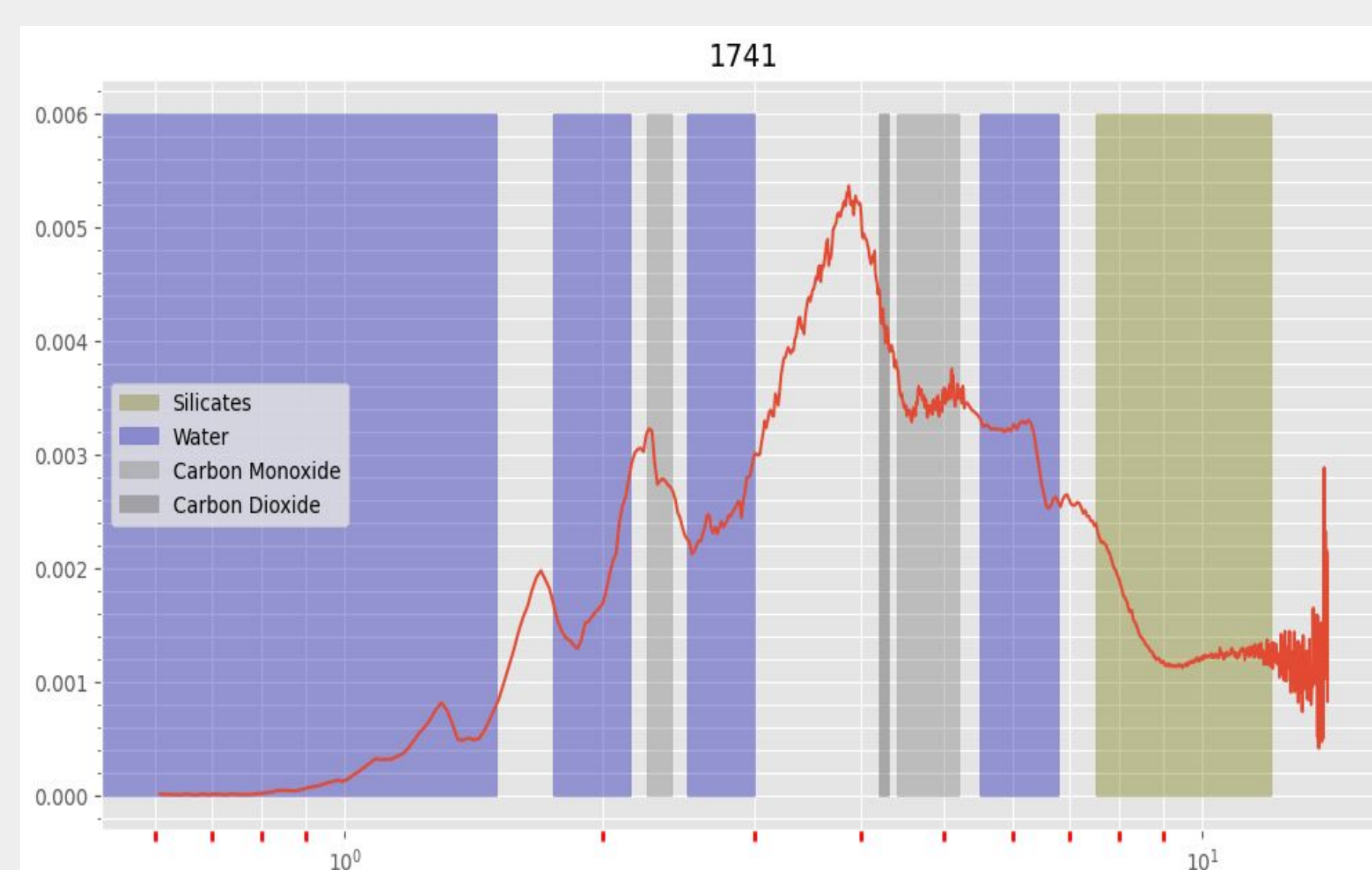
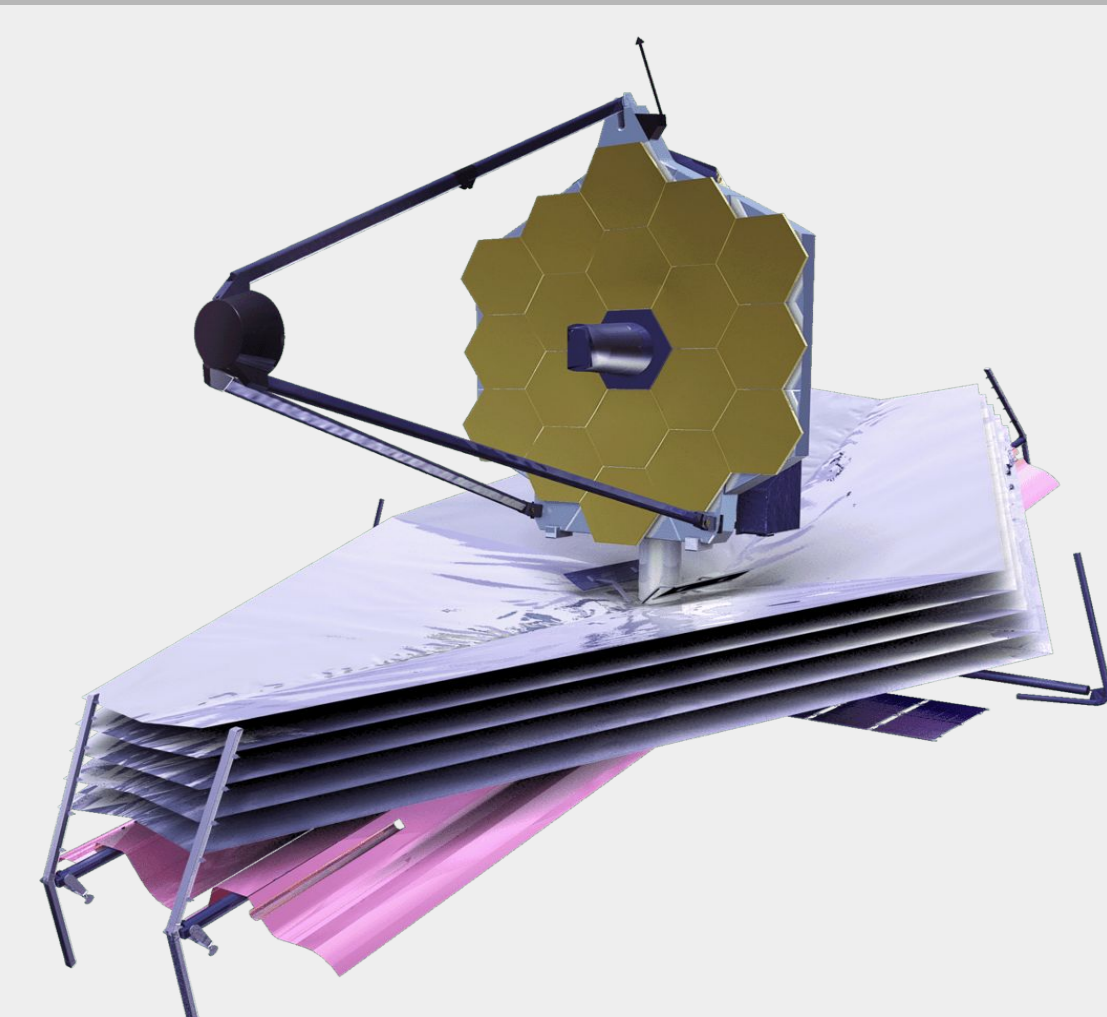


Figure 1: Example of layer absorption and a spectral plot.

Data

We analyzed spectra from six brown dwarfs collected by NASA's James Webb Space Telescope (JWST), published in Lam et al. (2026). The spectra consisted of *infrared* wavelengths (Figure 1), which is light beyond what human eyes can see, so a telescope is needed.



James Webb Space Telescope

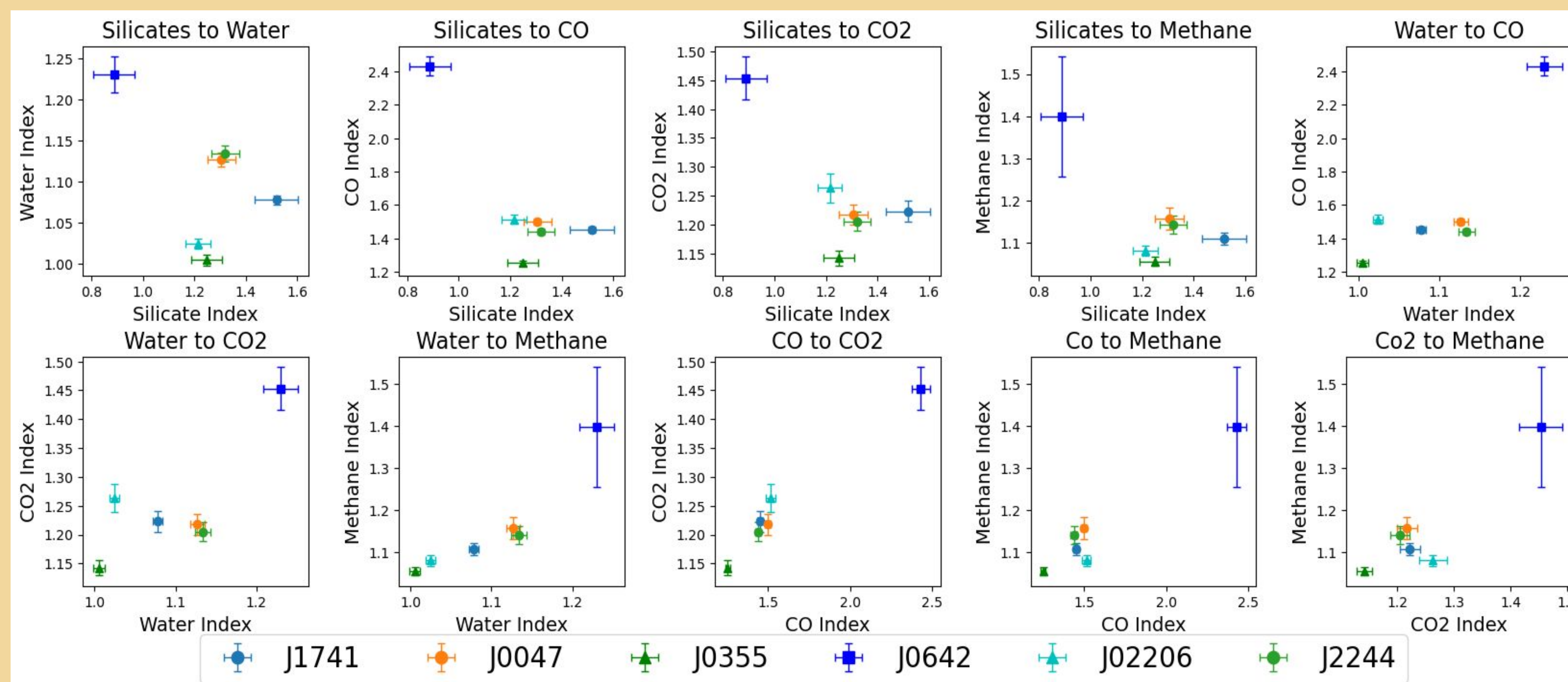


Figure 2: Comparison of spectral indices against each other.

Results

We identified the main features in the spectra at different wavelengths corresponding to: water at 1–3 microns (near-infrared) and 6–7 microns (mid-infrared), CH_4 at 3.3 microns, CO_2 at 4.3 microns, CO at 4.7 microns, and silicates around 9–12 microns (e.g., Kirkpatrick 2005). We measured the spectral indices for these molecules within each brown dwarf, represented by individual points in Figures 2 and 3.

- ❖ **We found that silicates have a mild negative correlation with all other molecules, particularly against methane.**
- ❖ Our results also show a slight positive correlation between H_2O in the near- and mid-infrared.

Discussion

These observations prove that silicates are a strong mid-infrared absorber, indicating that all of these brown dwarfs contain sand clouds in their atmospheres. **Our results suggest that in atmospheres with more sand, we can detect fewer traces of other molecules.** However, with non-silicate molecules the opposite occurs, meaning that the increased presence of one molecule is a good indicator for the presence of the others. These relationships could be explained because silicates are solid molecules that block light from deeper layers, while the others are gas molecules that do not have the same effect.

Acknowledgements

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References

Kirkpatrick 2005, ARAA, 43, 195
 Lam et al. 2026, A&A, accepted
 Suárez et al. 2026, JOSS, submitted

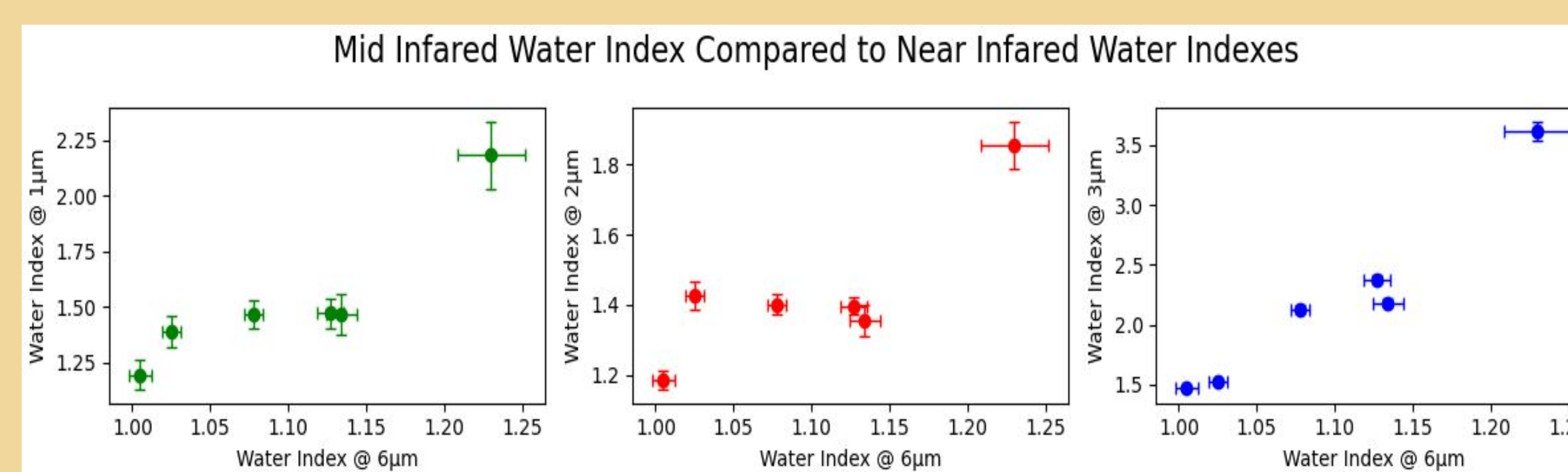


Figure 3: Comparing Mid Infrared Water index to Near Infrared indexes.